MAT 2040 Linear Algebra

Assignment 6

Please note that

- Released date: 2024/07/01
- Due: 2024/07/10
- Late submission is NOT acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW6".

Question 1. For each of the following, compute (i) $\det(\boldsymbol{A})$, (ii) adj $\boldsymbol{A}$, and (iii) $\boldsymbol{A}^{-1}$:

(a). $\boldsymbol{A} = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$ (b). $\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Question 2. Use Cramer's rule to solve each of the following systems:

(a). $\begin{cases} x_1 + 2x_2 = 3 \\ 3x_1 - x_2 = 1 \end{cases}$. (b). $\begin{cases} x_1 + 3x_2 + x_3 = 1 \\ 2x_1 + x_2 + x_3 = 5 \\ -2x_1 + 2x_2 - x_3 = -8 \end{cases}$.

Question 3. Let

$$\boldsymbol{P} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}, \quad \boldsymbol{\Lambda} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad \boldsymbol{AP} = \boldsymbol{P\Lambda}.$$

Find $\boldsymbol{A}^n$.

Question 4. Show that if $\boldsymbol{A}$ is nonsingular, then $\text{adj}(\boldsymbol{A})$ is nonsingular and

$$(\text{adj}\, \boldsymbol{A})^{-1} = \det(\boldsymbol{A}^{-1})\boldsymbol{A} = \text{adj}(\boldsymbol{A}^{-1}).$$

Question 5. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be 3×3 matrices, with $\det(\boldsymbol{A}) = -3$ and $\det(\boldsymbol{B}) = -4$. Use the properties of determinants to compute:

(a). $\det(\boldsymbol{AB})$; (b). $\det(5\boldsymbol{A})$; (c). $\det(\boldsymbol{B}^T)$; (d). $\det(\boldsymbol{A}^{-1})$; (e). $\det(\boldsymbol{A}^3)$.

Question 6. Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be 2×2 matrices and let

$$\boldsymbol{C} = \begin{bmatrix} a_{11} & a_{12} \\ b_{21} & b_{22} \end{bmatrix}, \qquad \boldsymbol{D} = \begin{bmatrix} b_{11} & b_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Show that

$$\det(\boldsymbol{A} + \boldsymbol{B}) = \det(\boldsymbol{A}) + \det(\boldsymbol{B}) + \det(\boldsymbol{C}) + \det(\boldsymbol{D}).$$

Question 7. Let U be a square matrix such that $U^T U = I$. Show that $\det(U) = \pm 1$.

Question 8. Assume matrix A is square, make each statement True or False

(a) If the columns of A are linearly dependent, then $\det(A) = 0$.

(b) $\det(A^{-1}) = (-1)\det(A)$.

(c) The determinant of A is the product of the diagonal entries in A.

(d) If $\det(A)$ is zero, then two rows or two columns are the same, or a row or a column is zero.

(e) $\det(A + B) = \det(A) + \det(B)$.

Question 9. Let

$$A = \begin{bmatrix} 1 & 5 & 2 \\ 2 & 4 & -1 \\ 1 & -2 & 1 \end{bmatrix}$$

(a) Find the values of $\det(M_{21})$, $\det(M_{22})$ and $\det(M_{23})$.

(b) Find the values of A_{21}, A_{22} and A_{23}.

(c) Use your answers from part (b) to compute $\det(A)$.

Question 10. Compute determinants using the cofactor formula.

(a) $\boldsymbol{A} = \begin{bmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{bmatrix}$ (b) $\boldsymbol{B} = \begin{bmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 2 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{bmatrix}$

Question 11. Let $L_1 : \mathcal{U} \to \mathcal{V}$ and $L_2 : \mathcal{V} \to \mathcal{W}$ be linear transformations, and let $L = L_2 \circ L_2$ be the mapping defined by

$$L(\boldsymbol{u}) = L_2(L_1(\boldsymbol{u}))$$

for each $\boldsymbol{u} \in \mathcal{U}$. Show that L is a linear transformation mapping $\mathcal{U}$ into $\mathcal{W}$.

Question 12. Consider the 3×3 Vandermonde matrix

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix}$$

(a) show that $\det(V) = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$.

(b) What conditions must the scalars x_1, x_2 and x_3 satisfy in order for V to be nonsingular?

Question 13. Let

$$\boldsymbol{b}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \boldsymbol{b}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad \boldsymbol{b}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix},$$

and let L be the linear transformation from $\mathbb{R}_2$ into $\mathbb{R}_3$ defined by

$$L(\boldsymbol{x}) = x_1\boldsymbol{b}_1 + x_2\boldsymbol{b}_2 + (x_1 + x_2)\boldsymbol{b}_3.$$

Find the matrix $\boldsymbol{A}$ representing L with respect to the standard basis $\{\boldsymbol{e}_1, \boldsymbol{e}_2\}$ of $\mathbb{R}_2$ and the basis $\{\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3\}$ of $\mathbb{R}_3$.

Question 14. Matrix A is called to be similar to matrix B if there is an invertible matrix S such that $A = S^{-1}BS$. Now prove the following properties of similarity:

(a) Any square matrix A is similar to itself.

(b) If B is similar to A, then A is similar to B.

(c) If A is similar to B and B is similar to C, then A is similar to C.

Question 15. Find the matrix representation of the linear transformation $T(f(t)) = f(3t - 2)$ from $\mathbb{P}_2$ to $\mathbb{P}_2$, where $\mathbb{P}_2 = \{a_0 + a_1x + a_2x^2 \mid a_0, a_1, a_2 \in \mathbb{R}\}$.

Question 16. Let L be the linear tranformation mapping $\mathbb{R}^3$ into $\mathbb{R}^3$ defined by $L(x) = Ax$, where

$$A = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & -2 \\ 2 & -1 & -1 \end{bmatrix}$$

and let

$$\boldsymbol{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \boldsymbol{v}_3 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

Find the transition matrix V corresponding to a change of basis from $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$ to $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$, and use it to determine the matrix B representing L with respect to $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\}$.

Question 17. A linear transformation $L : V \to W$ is said to be one-to-one if $L(\mathbf{v}_1) = L(\mathbf{v}_2)$ implies that $\mathbf{v}_1 = \mathbf{v}_2$ (i.e., no two distinct vectors $\mathbf{v}_1, \mathbf{v}_2$ in V get mapped into the same vector $\mathbf{w} \in W$. Show that L is one-to-one if and only if $\ker(L) = \{0_V\}$.

Question 18. For the following linear transformations $T : \mathbb{R}^n \to \mathbb{R}^n$, find a matrix A such that $T(\vec{x}) = A\vec{x}$ for all $\vec{x} \in \mathbb{R}^n$.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^3$,

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x - y \\ 3y \\ 4x + 5y \end{bmatrix}$$

(b) $T : \mathbb{R}^2 \to \mathbb{R}^2$, satisfying

$$T\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, T\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 \\ 5 \end{bmatrix}$$

Question 19. Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \vec{v}_1 = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$, and $\vec{v}_2 = \begin{bmatrix} 7 \\ -2 \end{bmatrix}$, and let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation that maps $\vec{x}$ into $x_1\vec{v}_1 + x_2\vec{v}_2$. Find a matrix A such that $T(\vec{x})$ is $A\vec{x}$ for each $\vec{x}$.

Question 20. Show that the transformation T defined by

$$T(x_1, x_2) = (x_1 - 2|x_2|, x_1 - 4x_2)$$

is not linear.